

Scattering from Elliptical Hair Fibers Based on Microfacet Theory

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Figure 1: Hair of different eccentricities under various illumination angles. Elliptical hair shows orientation-dependent glints.

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1 INTRODUCTION

Almost all hair and fiber scattering models in computer graphics today are based on separable *bidirectional curve scattering distribution function* (BCSDF) proposed by [Marschner et al. 2003], including the most recent works that model elliptical hair fibers [Benamira and Pattanaik 2021; Khungurn and Marschner 2017]; they separate the scattering function into a longitudinal function $M(\theta)$ and an azimuthal function $N(\varphi)$, which is mathematically convenient, but not physically correct, hence fails to capture the glinty appearance of real hair in the forward scattering direction.

We introduce an alternative scattering model for elliptical hair [Huang et al. 2022]. We consider the hair fiber as a homogeneous dielectric cylinder with a rough surface modeled as microfacets [Heitz 2014], and perform the usual radiometric analysis as by

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[Cook and Torrance 1982]. The resulting expression for the R lobe (reflection off the cylinder surface) turns out to be very intuitive; secondary lobes are more complicated due to multiple interactions with the surface, but can still be evaluated on the fly.

2 MICROFACET-BASED BCSDF

Microfacet theory models a rough, flat surface with macronormal ω_m as a collection of facets with micronormal ω_h , obeying *normal distribution function* (NDF) $D(\omega_h)$. The *bidirectional reflectance distribution function* (BRDF) of such a surface is

$$f_r(\omega_i, \omega_o) = \frac{R(\omega_h, \omega_o)G(\omega_o, \omega_i)D(\omega_h)}{4|\omega_m \cdot \omega_o||\omega_m \cdot \omega_i|}, \quad (1)$$

with $\omega_{i,o}$ being the incoming and outgoing direction, respectively, $R(\omega_h, \omega_o)$ the Fresnel reflectance, and the geometry term $G(\omega_o, \omega_i)$ models the mutual shadowing and masking between the microfacets. For hair, the surface is not flat, but an elliptical cylinder (Fig. 2). It turns out that one needs only to integrate the NDF along the azimuth to obtain the R lobe of the far-field BCSDF

$$S_R(\omega_i, \omega_o) = \frac{R_1}{4d_o \cos \theta_o \cos \theta_i} \int D_1 G_1 B(\gamma_{m1}) d\gamma_{m1}, \quad (2)$$

where $d_o = 2\sqrt{a^2 \cos^2 \varphi_o + b^2 \sin^2 \varphi_o}$ is the projected diameter viewing from ω_o , and $B(\gamma) = \sqrt{a^2 \cos^2 \gamma + b^2 \sin^2 \gamma}$ the arc length. Scale tilt α is modeled as a tilt of ω_m from the $x-z$ plane. For TT

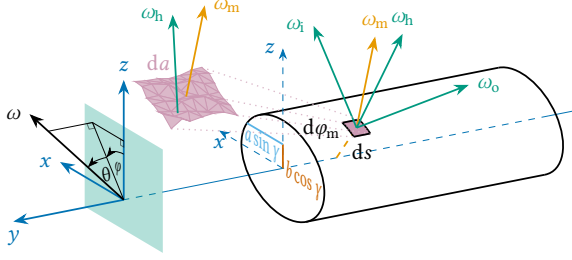


Figure 2: An illustration of the geometry and the coordinate system of our model. The x, z axes are aligned with the semi-major and the semi-minor axis, respectively.

and TRT lobes, one also needs to integrate along the internal path

$$S_{TT}(\omega_i, \omega_o) = \frac{1}{d_o \cos \theta_o \cos \theta_i} \iint \frac{T_1 T_2 D_1 D_2 G_1 G_2 A_t B(\gamma_{m2})}{\|\bar{\omega}_{h1}\|^2 \|\bar{\omega}_{h2}\|^2 |\omega_t \cdot \omega_{m1}|} \quad (3)$$

$$|\omega_i \cdot \omega_{h1}| |\omega_t \cdot \omega_{h1}| |\omega_o \cdot \omega_{h2}| |\omega_t \cdot \omega_{h2}| d\gamma_{m1} d\omega_t,$$

$$S_{TRT}(\omega_i, \omega_o) = \frac{1}{4d_o \cos \theta_o \cos \theta_i} \iiint \frac{T_1 R_2 T_3 D_1 D_2 D_3 G_1 G_2 G_3}{\|\bar{\omega}_{h1}\|^2 \|\bar{\omega}_{h3}\|^2 |\omega_t \cdot \omega_{m1}|} \frac{A_{trt} B(\gamma_{m3})}{|\omega_{tr} \cdot \omega_{m2}|} d\gamma_{m1} d\omega_{tr} d\omega_t,$$

where $\omega_{mi}, \omega_{hi}, T_i, R_i, D_i, G_i$ are the macronormals, micronormals, transmittance, reflectance, NDFs and geometry terms at interface $\textcircled{i}$, respectively (Fig. 3), and $A_{t, trt}$ are internal absorption. $\bar{\omega}_{h1} = -\omega_i - \eta \omega_t, \bar{\omega}_{h2} = -\omega_t + \omega_o/\eta$ and $\bar{\omega}_{h3} = \omega_{tr} + \omega_o/\eta$, with η being the relative refractive index of hair to air.

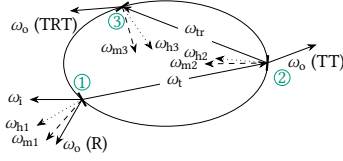


Figure 3: Internal scattering

3 IMPLEMENTATION

The hair vertices are taken from [Yuksel 2020b], based on which we form interpolating Bézier curves using the method in [Yuksel 2020a]. The hair geometry is modeled as flat stripes, with the semi-minor axis of the ellipse aligned with the curvature vector.

Higher-order integrations in Eq. (3) are evaluated via sampling an internal path based on the microfacet distribution, then dividing S by the probability of sampling that path, so that the resulting equations only integrate in γ_{m1} as in Eq. (2), which is then evaluated using numerical quadrature as in [d'Eon et al. 2011].

For importance sampling, we randomly sample a micronormal at each interface, then pick one of the three lobes proportional to their energy. The intersection between ω_i and the ellipse is given by $\gamma_{m1} = \text{atan2}(a \cos \phi_i, -b \sin \phi_i) - \cos^{-1}(2h/d_i)$, with $d_i = 2\sqrt{a^2 \cos^2 \phi_i + b^2 \sin^2 \phi_i}$ and $h \in (-\frac{d_o}{2}, \frac{d_o}{2})$ the azimuthal offset.

4 RESULTS

We compare the reflectance from a circular cross section (eccentricity $e = 0$) and an elliptical cross section with $e = 0.8$. The area of the two cross sections are the same.

A set of BCSDf slices for fixed incoming directions are visualized in Fig. 4. It is clear that the scattering lobes are not separable in θ_o

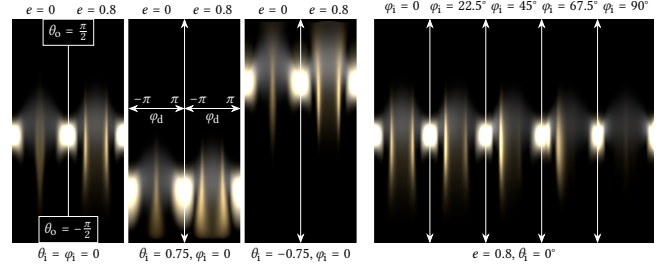


Figure 4: BCSDf slices as a function of θ_o and $\phi_d = \phi_o - \phi_i$, with $\alpha = 3^\circ$ and Beckmann roughness $\beta = 0.135$. *Left*: eccentricity $e > 0$ causes drastic changes in the TRT lobe. *Right*: reflectance from an elliptical cross section under various ϕ_i .

and ϕ_o . For $e = 0$, the TRT lobe has a highlight at $\phi_d = 0$; therefore, the hair looks bright when the camera and the illumination direction coincide (Fig. 1), and becomes darker under oblique illumination in ϕ_i . For $e = 0.8$, however, the TRT lobe usually has two widely separated highlights; therefore, the hair looks darker at normal illumination, but shows glints at oblique illumination angles which are much brighter than that of the circular hair fibers. The intensity of the glints also depends on ϕ_i (Fig. 4 right).

The images in Fig. 1 are rendered in 256 samples per pixel and have an initial resolution of 1024×768 ; on an Intel 12-core 3.3GHz i7-5820K CPU, they took around 40 min to render, about the same rendering time as [d'Eon et al. 2011].

5 DISCUSSION

We have demonstrated a physically-based scattering model for elliptical hair fibers. It is conceptually simple, efficient to evaluate and shows special characteristics such as multiple highlights in TRT lobe. Because the projected diameter depends on the viewing angle, our BCSDf is not reciprocal; therefore, special care should be taken for bidirectional path tracing. In Mitsuba 2 where ray is traced from the camera, we exchange ω_i and ω_o . We expect this model to also work for irregular cross sections.

REFERENCES

- Alexis Benamira and Sumanta Pattanaik. 2021. A Combined Scattering and Diffraction Model for Elliptical Hair Rendering. *Computer Graphics Forum* (2021). <https://doi.org/10.1111/cgf.14349>
- Robert L Cook and Kenneth E. Torrance. 1982. A reflectance model for computer graphics. *ACM Transactions on Graphics (TOG)* 1, 1 (1982), 7–24. <https://doi.org/10.1145/357290.357293>
- Eugene d'Eon, Guillaume Francois, Martin Hill, Joe Letteri, and Jean-Marie Aubry. 2011. An energy-conserving hair reflectance model. In *Computer Graphics Forum*, Vol. 30. Wiley Online Library, 1181–1187. <https://doi.org/10.1111/j.1467-8659.2011.01976.x>
- Eric Heitz. 2014. Understanding the Masking-Shadowing Function in Microfacet-Based BRDFs. *Journal of Computer Graphics Techniques (JCGT)* 3, 2 (30 June 2014), 48–107. <http://jcgt.org/published/0003/02/03/>
- Weizhen Huang, Matthias B Hullin, and Johannes Hanika. 2022. A Microfacet-based Hair Scattering Model. In *Computer Graphics Forum*, Vol. 41.
- Pramook Khungurn and Steve Marschner. 2017. Azimuthal scattering from elliptical hair fibers. *ACM Transactions on Graphics (TOG)* 36, 2 (2017), 1–23. <https://doi.org/10.1145/2998578>
- Stephen R Marschner, Henrik Wann Jensen, Mike Cammarano, Steve Worley, and Pat Hanrahan. 2003. Light scattering from human hair fibers. *ACM Transactions on Graphics (TOG)* 22, 3 (2003), 780–791. <https://doi.org/10.1145/882262.882345>
- Cem Yuksel. 2020a. A class of C^2 interpolating splines. *ACM Transactions on Graphics (TOG)* 39, 5 (2020), 1–14. <https://doi.org/10.1145/3400301>
- Cem Yuksel. 2020b. HAIR Model Files. www.cemyuksel.com/research/hairmodels